

Chapter # 13

Partial Derivatives

Exercise 13.1 Questions 1-8

Functions of Two or More Variables

Question NO 1:

Let $f(x, y) = x^2y + 1$. Find.

Solution.

$$a) f(2, 1) = (2)^2(1) + 1 = 4 + 1 = 5$$

$$b) f(1, 2) = (1)^2(2) + 1 = 2 + 1 = 3$$

$$c) f(0, 0) = (0)^2(0) + 1 = 0 + 1 = 1$$

$$d) f(1, -3) = (1)^2(-3) + 1 = -3 + 1 = -2$$

$$e) f(3a, a) = (3a)^2(a) + 1 = 9a^3 + 1 = 9a^3 + 1$$

$$f) f(ab, a-b) = (ab)^2(a-b) + 1 = a^2b^2(a-b) + 1 = a^3b^2 - a^2b^2 + 1$$

Question NO 2:

Let $f(x, y) = x + 3\sqrt[3]{xy}$. Find

Solution

$$a) f(t, t^2) = t + 3\sqrt[3]{(t)(t^2)} = t + 3\sqrt[3]{t^3} = t + (t^3)^{1/3} = 2t$$

$$b) f(x, x^2) = x + 3\sqrt[3]{(x)(x^2)} = x + 3\sqrt[3]{x^3} = x + (x^3)^{1/3} = 2x$$

$$c) f(2y^2, 4y) = 2y^2 + 3\sqrt[3]{(2y^2)(4y)} = 2y^2 + (8y^3)^{1/3} = 2y^2 + 3\sqrt[3]{8y^3} = 2y^2 + ((2y)^3)^{1/3}$$

Question No 3

Let $f(x, y) = xy + 3$. Find.

Solution:

$$\begin{aligned} \text{a) } f(x+y, x-y) &= (x+y)(x-y) + 3 = x^2 - y^2 + 3 \end{aligned}$$

$$\begin{aligned} \text{b) } f(xy, 3x^2y^3) &= (xy)(3x^2y^3) + 3 = 3x^3y^4 + 3 = 3(x^3y^4 + 1) \end{aligned}$$

Question No 4

Let $g(x) = x \sin x$. Find

Solution:

$$\begin{aligned} \text{a) } g\left(\frac{x}{y}\right) &= \frac{x}{y} \sin\left(\frac{x}{y}\right) \\ \text{b) } g(xy) &= xy \sin(xy) \end{aligned}$$

$$\text{c) } f(x-y) = (x-y) \sin(x-y)$$

Question No 5

Find $F(g(x), h(y))$ if $f(x, y) = xe^{xy}$,
 $g(x) = x^3$, and $h(y) = 3y+1$

$$F(g(x), h(y)) \text{ if } f(x, y) = xe^{xy}$$

$$f(g, h) = ge^{gh} = x^3 e^{x^3(3y+1)}$$

Question No 6:

Find $g(u(x,y), v(x,y))$ if $g(x,y) = y \sin(x^2 y)$, $u(x,y) = x^2 y^3$, and $v(x,y) = \pi x y$

$$g(u, v) = v \sin(u^2 v) = \pi x y \sin((x^2 y^3)^2 (\pi x y))$$

$$g(u, v) = \pi x y \sin(x^5 y^7 \pi)$$

Question No 7

Let $f(x,y) = x + 3x^2 y^2$, $x(t) = t^2$, and $y(t) = t^3$. Find.

$$\begin{aligned} \text{a) } f(x(t), y(t)) &= x + 3x^2 y^2 = t^2 + 3t^4 t^6 \\ f(x, y) &= t^2 + 3t^{10} \end{aligned}$$

$$\begin{aligned} \text{b) } f(x(0), y(0)) &= x(0) = (0)^2, \quad y(0) = (0)^3 \\ f(x, y) &= (0)^4 + 3(0)^2(0)^2 = 0 \end{aligned}$$

$$\begin{aligned} \text{c) } f(x(2), y(2)) \quad x(2) &= (2)^2 = 4 \\ y(2) &= (2)^3 = 8 \end{aligned}$$

$$f(x, y) = 4 + 3(4)^2(8)^3 = 4 + 3(16)(64)$$

$$f(x, y) = 4 + 3072$$

$$f(x, y) = 3076$$

Question NO 8.

Let $g(x, y) = ye^{-3x}$, $x(t) = \ln(t^2 + 1)$,
and $y(t) = \sqrt{t}$. Find $g(x(t), y(t))$

$$g(x(t), y(t)) = ye^{-3x(t)}$$

$$g(x(t), y(t)) = (\sqrt{t})e^{-3(\ln(t^2+1))}$$

$$\because -3 \ln x = \ln x^{-3}$$

$$g(x(t), y(t)) = \sqrt{t} e^{-3 \ln(t^2+1)}$$

$$g(x(t), y(t)) = \sqrt{t} e^{\ln(t^2+1)^{-3}}$$

$$g(x(t), y(t)) = \frac{\sqrt{t}}{(t^2+1)^3}$$

Question NO 17 Functions of 3 variables

Let $f(x, y, z) = xy^2z^3 + 3$. Find

$$\begin{aligned} \text{a) } f(2, 1, 2) &= (2)(1)^2(2)^3 + 3 = (2)(8) + 3 \\ &= 16 + 3 = 19 \end{aligned}$$

$$\begin{aligned} \text{b) } f(-3, 2, 1) &= (-3)(2)^2(1)^3 + 3 = (-3)(4)(1) + 3 \\ &= -12 + 3 = -9 \end{aligned}$$

$$c) f(0,0,0) = (0)(0)^2(0)^3 + 3 = 0 + 3 = 3$$

$$d) f(a,a,a) = (a)(a)^2(a)^3 + 3 = a^6 + 3$$

$$e) f(t, t^2, -t) = (t)(t^2)^2(-t)^3 + 3 = -t^8 + 3$$

$$f) f(a+b, a-b, b) = (a+b)(a-b)^2(b)^3 + 3$$

$$= (a+b)(a+b)(a-b)b^3 + 3$$

$$= (a^2 - b^2)(a-b)b^3 + 3$$

$$= [a^2(a-b) - b^2(a-b)]b^3 + 3$$

$$= [a^3 - a^2b - ab^2 + b^3]b^3 + 3$$

$$= a^3b^3 - a^2b^4 - ab^5 + b^6 + 3$$

Question No 18

Let $f(x,y,z) = 2xy + x$. Find

$$a) f(x+y, x-y, x^2) \quad b) f(xy, y/x, xz)$$

$$= x^2(x+y)(x-y) + x+y$$

$$= x^2(x^2 - y^2) + x+y$$

$$= x^4 - x^2y^2 + x+y$$

$$= xz(xy)(y/x) + xy$$

$$= xz(y^2 + xy)$$

$$= xy^2z + xy$$

Question No 19

Find $F(f(x), g(y), h(z))$ if $F(x, y, z) = ye^{xyz}$, $f(x) = x^2$, $g(y) = y+1$ and $h(z) = z^2$.

$$F(f(x), g(y), h(z)) = F(x^2, y+1, z^2)$$

$$F(x, y, z) = ye^{xyz}$$

$$F(x, y, z) = (y+1)e^{x^2(y+1)z^2}$$

$$F(f, g, h) = (y+1)e^{x^2z^2(y+1)}$$

Question No 20:

Find $g(u(x, y, z), v(x, y, z), w(x, y, z))$ if $g(x, y, z) = z \sin xy$, $u(x, y, z) = x^2z^3$, $v(x, y, z) = \pi xyz$ and $w(x, y, z) = xy/z$

$$g(\underset{\substack{\downarrow \\ \text{replace}}}{u}, \underset{\substack{\downarrow \\ \text{replace}}}{v}, \underset{\substack{\downarrow \\ \text{replace}}}{w}) = \underset{\substack{\downarrow \\ \text{replace}}}{x^2z^3} \sin \underset{\substack{\downarrow \\ \text{replace}}}{\pi xyz} \underset{\substack{\downarrow \\ \text{replace}}}{xy/z}$$

$$g(u, v, w) = z \sin xy = xy/z \sin x^2z^3(\pi xyz) \\ = \frac{xy}{z} \sin x^3y z^4 \pi$$

Exercise # 13.2

1-6 Use Limit Laws and continuity properties to evaluate the limit.

1: $\lim_{(x,y) \rightarrow (1,3)} (4xy^2 - x)$

By using Limit Law's

$$= 4 \lim (xy^2) - \lim (x)$$

$$= 4(1)(3)^2 - (1)$$

$$= 4 \times 9 - 1$$

$$= 36 - 1$$

$$= 35$$

2: $\lim_{(x,y) \rightarrow (0,0)} \frac{4x - y}{\sin y - 1}$

By using Limit Law's

$$= \frac{4 \lim (x) - \lim y}{\lim (\sin y) - 1}$$

$$= \frac{4(0) - (0)}{(0) - 1} = \frac{0}{-1}$$

$$= 0$$

3: $\lim_{(x,y) \rightarrow (-1,2)} \frac{xy^3}{x+y}$

By using Limit Law's

$$= \frac{\lim (xy^3)}{\lim (x) + \lim (y)} = \frac{\lim (-1)(2)^3}{\lim -1 + \lim 2} = \frac{-8}{1} = -8$$

$$4: \lim_{(x,y) \rightarrow (1,-3)} (e^{2x-y^2})$$

$$5: \lim_{(x,y) \rightarrow (0,0)} \ln(1+x^2y^3)$$

By using Limit Law's

$$= e^{2(1)-(-3)^2}$$

$$= e^{2-9}$$

$$= e^{-7}$$

By using Limit Law's

$$= \ln(1 + \lim x^2y^3)$$

$$= \ln(1 + (0)^2(0)^3)$$

$$= \ln(1)$$

$$= 0$$

$$6: \lim_{(x,y) \rightarrow (4,-2)} \sqrt[3]{y^3 + 2x}$$

By using Limit Law's

$$= \lim \sqrt[3]{\lim y^3 + 2 \lim x}$$

$$= \lim (4)^3 \sqrt[3]{(-2)^3 + 2(4)}$$

$$= \lim 4^3 \sqrt[3]{-\cancel{8} + \cancel{8}} = \lim 4^3 \sqrt[3]{0}$$

$$= \lim 4^3 \sqrt[3]{0} = 0$$

7-8 show that the limit does not exist by considering the limits as $(x,y) \rightarrow (0,0)$ along the coordinate axes.

$$7: (a) \lim_{(x,y) \rightarrow (0,0)} \frac{3}{x^2 + 2y^2}$$

we use along different curves.

Along x-axis

$$x = t \quad y = 0$$

$$= \lim_{t \rightarrow 0} \frac{3}{(t)^2 + 2(0)^2}$$

$$= \lim_{t \rightarrow 0} \frac{3}{t^2 + 0}$$

$$= \frac{3}{0+0} = \frac{3}{0} = \infty \text{ does not exist}$$

$$b) \lim_{(x,y) \rightarrow (0,0)} \frac{x+y}{2x^2+y^2}$$

we use along different curves.

Along x-axis

$$x = t \quad y = 0$$

$$= \lim_{t \rightarrow 0} \frac{t + 0}{2(t)^2 + (0)^2}$$

$$= \lim_{t \rightarrow 0} \frac{t}{2t^2 + 0}$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{t}{2tx}$$

$$= \lim_{t \rightarrow 0} \frac{1}{2(0)^2} = \frac{1}{0} = \infty \text{ does not exist}$$

8: (a) $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x^2+y^2}$

Along y-axis

$y = t \quad x = 0$

$$= \lim_{t \rightarrow 0} \frac{0 - t}{(0)^2 + (t)^2}$$

$$= \lim_{t \rightarrow 0} \frac{-t}{t^2}$$

$$= \lim_{t \rightarrow 0} \frac{1}{0} = \infty \text{ does not exist}$$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{\cos xy}{x^2+y^2}$

Along x-axis

$x = t, y = 0$

$$= \lim_{t \rightarrow 0} \frac{\cos(t) \cdot (0)}{(t)^2 + (0)^2}$$

$$= \lim_{t \rightarrow 0} \frac{\cos t \cdot 0}{t^2 + 0}$$

$$= \lim_{t \rightarrow 0} \frac{\cos 0}{t} = \frac{1}{t} = \frac{1}{0} = \infty$$

does not exist

9-12 Evaluate the limit using the substitution $z = x^2 + y^2$ and observing that $z \rightarrow 0^+$ if and only if $(x, y) \rightarrow (0, 0)$
Hospital Rule

$$9: \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} \quad (x,y) \rightarrow (0,0)$$

$$\text{Let } z = x^2 + y^2 \quad z \rightarrow 0^+$$

$$\text{Then, } \lim_{z \rightarrow 0^+} \frac{\sin(z)}{z}$$

$$\because \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$= 1$$

$$10: \lim_{(x,y) \rightarrow (0,0)} \frac{1 - \cos(x^2 + y^2)}{x^2 + y^2}$$

$$\text{Let } z = \frac{x^2 + y^2}{x^2 + y^2} \quad \text{if and only if } (x,y) \rightarrow (0,0)$$

$$= \lim_{z \rightarrow 0^+} \frac{1 - \cos z}{z} = \left(\frac{0}{0} \text{ form} \right) \quad z \rightarrow 0^+ \quad x^2 + y^2 = z^2$$

$$x^2 + y^2 = z$$

indeterminate
undefined function

Apply Hospital Rule

$$= \lim_{z \rightarrow 0^+} \frac{\sin z}{1} = \sin(0) = 0$$

$$11: \lim_{(x,y) \rightarrow (0,0)} e^{-1/(x^2+y^2)}$$

$$\text{let } Z = x^2 + y^2 \quad \because (e^\infty = \infty)$$

$$= \lim_{Z \rightarrow 0^+} e^{-1/Z}$$

$$= \lim_{Z \rightarrow 0^+} e^{-1/0} = e^{-\infty} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$$

$$12: \lim_{(x,y) \rightarrow (0,0)} \frac{e^{-1/\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}$$

$$\text{let } Z = x^2 + y^2$$

$$= \lim_{Z \rightarrow 0^+} \frac{e^{-1/\sqrt{Z}}}{\sqrt{Z}}$$

$$= \lim_{Z \rightarrow 0^+} \frac{1}{\sqrt{Z}} e^{-1/\sqrt{Z}}$$

$$\text{let } w = 1/\sqrt{Z} \text{ put}$$

$$= \lim_{w \rightarrow +\infty} w \cdot e^{-w}$$

$$= \lim_{w \rightarrow +\infty} \frac{w}{e^w} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$$

By hospital Rule

$$= \lim_{w \rightarrow +\infty} \frac{1}{e^w} = \frac{1}{\infty} = 0$$

13-22. Determine whether the limit exists. if so, find its value.

$$13 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^4 + y^4}{x^2 + y^2} \quad \left(\frac{0}{0} \text{ form}\right)$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{\cancel{(x^2 + y^2)} (x^2 - y^2)}{\cancel{x^2 + y^2}}$$

$$= \lim_{(x,y) \rightarrow (0,0)} (x^2 - y^2)$$

$$= (0)^2 - (0)^2 = 0 - 0 = 0$$

$$14: \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 16y^2}{x^2 + 4y^2} \quad \left(\frac{0}{0} \text{ form}\right)$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{\cancel{(x^2 + 4y^2)} (x^2 - 4y^2)}{\cancel{x^2 + 4y^2}}$$

$$= \lim_{(x,y) \rightarrow (0,0)} x^2 - 4y^2$$

$$= (0)^2 - 4(0)^2$$

$$= 0 - 0 = 0$$

$$15: \quad \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{3x^2 + 2y^2} \quad \left(\frac{0}{0} \text{ form}\right)$$

we use along different Curves.

(i) Along x-axis

$$x = t \quad y = 0$$

$$= \lim_{t \rightarrow 0} \frac{t(0)}{3(t)^2 + 2(0)^2} = \lim_{t \rightarrow 0} \frac{0}{3t^2 + 0}$$

$$= \lim_{t \rightarrow 0} \frac{0}{3t^2} = 0$$

(ii) Along y=x so that,

$$x = t \quad y = t$$

$$= \lim_{t \rightarrow 0} \frac{t \cdot t}{3(t)^2 + 2(t)^2} = \lim_{t \rightarrow 0} \frac{t^2}{3t^2 + 2t^2}$$

$$= \lim_{t \rightarrow 0} \frac{t^2}{5t^2} = \frac{1}{5} \quad \text{limit does not exist}$$

16: $\lim_{(x,y) \rightarrow (0,0)} \frac{1 - x^2 - y^2}{x^2 + y^2}$

$$\text{let } z = x^2 + y^2$$

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{1 - (x^2 + y^2)}{x^2 + y^2}$$

$$= \lim_{z \rightarrow 0} \frac{1 - z}{z}$$

$$= \frac{1 - 0}{0} = \frac{1}{0} = \infty \quad \text{does not exist}$$

$$17: \lim_{(x,y,z) \rightarrow (2,-1,2)} \frac{xz^2}{\sqrt{x^2+y^2+z^2}}$$

$$\lim_{(x,y,z) \rightarrow (2,-1,2)} \frac{xz^2}{\sqrt{x^2+y^2+z^2}}$$

$$= \frac{(2)(2)^2}{\sqrt{(2)^2 + (-1)^2 + (2)^2}}$$

$$= \frac{8}{\sqrt{4+1+4}} = \frac{8}{\sqrt{9}} = \frac{8}{3}$$

$$18: \lim_{(x,y,z) \rightarrow (2,0,-1)} \ln(2x+y-z)$$

$$= \lim_{(x,y,z) \rightarrow (2,0,-1)} \ln(2x+y-z)$$

$$= \ln(2(2) + 0 - (-1))$$

$$= \ln(4 + 0 + 1)$$

$$= \ln(5)$$

$$= \ln 5$$

$$19: \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin(x^2+y^2+z^2)}{\sqrt{x^2+y^2+z^2}}$$

$$\text{Let } t = \sqrt{x^2 + y^2 + z^2}$$

$$(x, y, z) \rightarrow (0, 0, 0)$$

$$t \rightarrow 0^+$$

$$\lim_{t \rightarrow 0^+} \frac{\sin t^2}{t}$$

By using L. Hospital Rule

$$\lim_{t \rightarrow 0^+} \frac{\cos t^2 \cdot 2t}{1} = \lim_{t \rightarrow 0^+} (\cos t^2)(2t)$$

$$= (\cos(0)^2)(2(0))$$

$$= \cos(0)(0) = (1)(0)$$

$$= 0$$

20: $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin \sqrt{x^2 + y^2 + z^2}}{x^2 + y^2 + z^2}$

$$\text{Let } t = \sqrt{x^2 + y^2 + z^2}$$

$$t = \frac{\sqrt{x^2 + y^2 + z^2}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\lim_{t \rightarrow 0^+} \frac{\sin t}{t^2}$$

By using hospital Rule

$$\lim_{t \rightarrow 0^+} \frac{\cos t}{2t} = \frac{\cos(0)}{2(0)} = \frac{1}{0} = \infty$$

$$21: \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{e^{\sqrt{x^2+y^2+z^2}}}{\sqrt{x^2+y^2+z^2}}$$

let

$$t = \sqrt{x^2+y^2+z^2}$$

$$(x,y,z) \rightarrow (0,0,0)$$

$$t \rightarrow 0^+$$

Then,

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{e^{\sqrt{x^2+y^2+z^2}}}{\sqrt{x^2+y^2+z^2}}$$

$$= \lim_{t \rightarrow 0^+} \frac{e^t}{t} = \frac{e^{(0)}}{0} = \frac{1}{0} = \infty$$

limit does not exist

$$22: \lim_{(x,y,z) \rightarrow (0,0,0)} \tan^{-1} \left(\frac{1}{x^2+y^2+z^2} \right)$$

let

$$t = x^2+y^2+z^2$$

$$(x,y,z) \rightarrow (0,0,0)$$

$$t \rightarrow 0^+$$

Then,

$$\lim_{t \rightarrow 0^+} \tan^{-1} \left(\frac{1}{t} \right)$$

$$= \tan^{-1} \left(\frac{1}{0} \right) = \tan^{-1}(\infty) = \frac{\pi}{2}$$

23-26: Evaluate the limit by
Converting to polar co-ordinate.

$$23: \lim_{x,y \rightarrow (0,0)} \sqrt{x^2+y^2} \ln(x^2+y^2)$$

Let (r, θ) be polar coordinate of
the point (x, y) with $r \geq 0$

$$x = r \cos \theta \rightarrow \textcircled{1} \quad y = r \sin \theta \rightarrow \textcircled{2}$$

Add eq. $\textcircled{1}$ with $\textcircled{2}$ and sq.

$$x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta$$

$$x^2 + y^2 = r^2 (\cos^2 \theta + \sin^2 \theta)$$

$$\boxed{x^2 + y^2 = r^2}$$

$$r = \sqrt{x^2 + y^2}$$

$$x, y \rightarrow (0, 0) \quad , \quad r \rightarrow 0^+$$

$$\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2+y^2} \ln(x^2+y^2)$$

$$\lim_{r \rightarrow 0^+} r \ln(r^2) \quad \because \begin{matrix} \ln x^n \\ n \ln x \end{matrix}$$

$$= \lim_{r \rightarrow 0^+} \frac{2 \ln r}{1/r} = \left(\frac{\infty}{\infty} \right) \text{ indeterminate form}$$

using L-Hospital Rule.

$$\lim_{r \rightarrow 0^+} \frac{2r}{-1/r^2}$$

$$\frac{r^1}{-1 \cdot r^{-1-1}}$$

$$\lim_{r \rightarrow 0^+} (-2r)$$

$$-1 \cdot r^{-2}$$

Apply limit

$$-2(0) = 0$$

24: $\lim_{(x,y) \rightarrow (0,0)} y \ln(x^2 + y^2)$

let (r, θ) be the polar co-ordinate at point of (x, y) with $r \geq 0$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$r = \sqrt{\quad}$$

$$r^2 = x^2 + y^2$$

$(x, y) \rightarrow (0, 0)$ if and only if $r \rightarrow 0^+$

$$\lim_{(x,y) \rightarrow (0,0)} y \ln(x^2 + y^2) = \lim_{r \rightarrow 0^+} r \sin \theta \ln r^2$$

$$= \lim_{r \rightarrow 0^+} r \sin \theta \cdot 2 \ln r$$

$$= \lim_{r \rightarrow 0^+} 2r (\ln r) \sin \theta$$

$$= 2(0)(\ln(0) \sin \theta)$$

$$= (0)(\infty)(\sin \theta)$$

$$= 0$$

$$25 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{\sqrt{x^2 + y^2}}$$

let (r, θ) be the polar coordinate at a point (x, y) with $r \geq 0$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$(x, y) \rightarrow (0, 0), \quad r \rightarrow 0^+$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{\sqrt{x^2 + y^2}} = \lim_{r \rightarrow 0^+} \frac{r^2 y^2}{\sqrt{x^2 + y^2}}$$

$$= \lim_{r \rightarrow 0^+} \frac{(r^2)(\cos^2 \theta) \cdot r^2 \sin^2 \theta}{r}$$

$$= \lim_{r \rightarrow 0^+} \frac{r^4 \cos^2 \theta \sin^2 \theta}{r}$$

$$= \lim_{r \rightarrow 0^+} (r^3 \cos^2 \theta \sin^2 \theta)$$

$$= (0)^3 \cos^2 \theta \cdot \sin^2 \theta$$

$$= 0 \cdot \cos^2 \theta \cdot \sin^2 \theta$$

$$= 0$$

$$26: \lim_{x,y \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + 2y^2}}$$

Let (r, θ) , be the polar coordinate at a point (x, y) with $r \geq 0$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$(x, y) \rightarrow (0, 0)$ if and only if $r \rightarrow 0^+$

$$= \lim_{r \rightarrow 0^+} \frac{r \cos \theta \cdot r \sin \theta}{\sqrt{(r \cos \theta)^2 + 2(r \sin \theta)^2}}$$

$$= \lim_{r \rightarrow 0^+} \frac{r^2 \cos \theta \cdot \sin \theta}{\sqrt{r^2 \cos^2 \theta + 2r^2 \sin^2 \theta}}$$

$$= \lim_{r \rightarrow 0^+} \frac{r^2 \cos \theta \cdot \sin \theta}{\sqrt{2} \cdot \sqrt{r^2} \sqrt{\cos^2 \theta + \sin^2 \theta}}$$

$$= \lim_{r \rightarrow 0^+} \frac{r^2 \cos \theta \cdot \sin \theta}{\sqrt{2} \cdot r \sqrt{1}}$$

$$= \lim_{r \rightarrow 0^+} \frac{r \cos \theta \cdot \sin \theta}{\sqrt{2}} = \frac{(0) \cos \theta \cdot \sin \theta}{\sqrt{2}}$$

$$= \frac{0}{\sqrt{2}} = 0$$

27-28: Evaluate the limits by converting to spherical coordinate (ρ, θ, ϕ) and by observing that $\rho \rightarrow 0^+$ if and only if $(x, y, z) \rightarrow (0, 0, 0)$

$$27. \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{xyz}{x^2+y^2+z^2}$$

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \\ z = \rho \cos \phi$$

$$\left| \frac{xyz}{x^2+y^2+z^2} \right| = \left| \frac{\rho^3 \sin^2 \phi \cos \phi \sin \theta \cos \theta}{\rho^2} \right| \leq \rho$$

$$\lim_{\rho \rightarrow 0^+} \rho = 0 \Rightarrow \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{xyz}{x^2+y^2+z^2} = 0$$

$$28. \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin x \sin y}{\sqrt{x^2+2y^2+3z^2}}$$

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

$$\left| \frac{\sin x \sin y}{\sqrt{x^2+2y^2+3z^2}} \right| \leq \left| \frac{xy}{\sqrt{x^2+y^2+z^2}} \right| = \frac{\rho^2 \sin^2 \phi \cos \theta \sin \theta}{\rho} \leq \rho$$

$$\lim_{\rho \rightarrow 0^+} \rho = 0 \Rightarrow \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin x \sin y}{\sqrt{x^2+2y^2+3z^2}} = 0$$